BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
# VITAMIN A (RETINOL)	0.5		mg /L	0.3 - 0.6	HPLC

INTERPRETATION NOTES :

Remark:

According to WHO guidelines a beginning deficiency in adults is indicated by concentrations between 0.1 to 0.19 mg/l, whereas concentrations < 0.1 mg/l indicate a minifest deficiency, Concentrations > 1.4 mg/l are considered as toxic.

Description Of Test

Vitamin A is the nutritional term for the group of compounds with a 20 carbon structure containing a methyl -substituted cyclohexenyl ring (Beta -ionone ring) & an isoprenoid side chain, with either a hydroxyl group (retinol),an aldehyde group (retinal), a carboxylic acid group(retinoic acid) or an ester group (retinyl ester) at the terminal C 15. Vitamin A deficiency causes degenerative changes in eyes & skin, & poor dark adaptation or night blindness (nyctalopia). More serious effects of deficiency are xerophthalmia, in which the conjunctiva becomes dry with small gray plaques with foamy surfaces (bitots spot), keratomalacia.The skin changes include dryness, roughness,papular eruption & follicular hyperkeratosis.

Decreased IN

1. Preterm infants specially infants with birth weight less than 1500g (< 30 weeks of gestation).
2. Nutritional deficiency
3. Fat malabsorption, particularly caused by celiac disease or chronic pancreatitis.
4. Hepatic disease or alcohol abuse.

Increased IN

Hypervitaminosis A results as a side effect of inappropriate therapy or ingestion of excess Vitamin A.

Sample Type : *EDTA plasma*

End of Report

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
M.D (Pathology)

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COLLEGE of AMERICAN PATHOLOGISTS

P.O Box: 49527 Dubai, UAE Tel: +971 4 398 8567 reports@biosytech.ae www.biosytech.com

BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
# VITAMIN B1 (THIAMIN)	72.69		ug/L	28 - 85	HPLC

INTERPRETATION NOTES :

• Vitamin B1 is increased in leukemia, polycythemia vera, and Hodgkin's disease.
• It is decreased in alcoholism with and without liver disease., in deficient diet, chronic febrile infections, prolonged diarrhoea, diabetes, carcinoid syndrome Hartnup disease, and pellagra.
Clinical utility:
• Measurement of Vitamin B1 is useful in the assessment of thiamine deficiency, in the investigation of suspected beriberi, and in monitoring the effects of chronic alcoholism.
Limitations:
• Drugs that may decrease vitamin B1 levels include glibenclamide, isoniazid, and valproic acid.
• Diets high in freshwater fish and tea, which are thiamine antagonists may cause decreased vitamin B1 levels.
Associated Tests:
• V0036-Vitamin B Complex profile, V0012-Vitamin B2 Riboflavin EDTA Whole Blood, V8123-Vitamin B3 (Niacin), V8125- Vitamin B5, V0013-Vitamin B6 Pyridoxine EDTA Whole Blood, V8124-Vitamin B7 (Biotin)
Reference:
• Kit Insert
• Wallach's Interpretation of Diagnostic tests-10th Edition

# VITAMIN B6 (PYRIDOXAL PHOSPHATE)	**66.00**	**H**	ng/mL	8.7 - 27.2	HPLC

Comments : Correlation with clinical history is suggested.

INTERPRETATION NOTES :

1. The main circulating forms of Vitamin B6 are pyridoxal phosphate (PLP), pyridoxal(PL) & pyridoxic acid(PA).
2. The active form in the body is PLP.It acts as a coenzyme for many enzymes in the body & is involved in metabolism of protein, carbohydrates & fats, production of Serotonin & haem & binding of steroid hormones to their receptors.
3. Dietary deficiency is uncommon. The common cause of deficiency is ingestion or use of antagonists like cyclosporine, hydrazaline, Isoniazide & penicillamine.
4. Sideroblastic anaemias have been described that are responsive to Vitamin B6 administration.
5. It may be decreased in pregnancy,lactation,alcoholism,diabetes mellitus, & in an uncommon dependency states.

RECOMMENDATIONS :

1. Withhold Vitamin B6 supplement for 24hrs before test.

REFERENCES :

1. Bender DA, Novel functions of Vitamin B6,Proc Nutr Soc,1994,53:625-30.
2. Borschel MW,Kirskey A, & Hannemann RE,Effects of Vitamin B6 intake on Growth of Young I infants, "Am J Clin Nutr,1986,43(1):7 -15.

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
M.D (Pathology)

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					

This test is performed in an accredited referral laboratory.

Sample Type : *EDTA Whole Blood*

End of Report

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
M.D (Pathology)

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
# Vitamin B12, active Holo Transcobalamin	107.90		pmol/L	25.1-165	CMIA

INTERPRETATION NOTES :

The Transcobolamin-Vitamin B12 Complex (Biologically avaliable cobalamin) is called Holotranscobalamin which promotes the uptake of its cobalamin by all cells.

This test is performed in an accredited referral laboratory.

Sample Type : *Serum*

End of Report

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M.D (Pathology)
Pathologist

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
# VITAMIN E	17.06		mg /L	5-18	HPLC

INTERPRETATION NOTES :

DESCRIPTION OF TEST :
Vitamin E is the nutritional term for the group of naturally occurring tocopherols & tocotrienols that have biological activity similar to RRR -alpha -tocopherol.

Vitamin E is considered necessary for (1) neurological & reproductive functions, (2) protecting the RBCs from hemolysis, (3) prevention of retinopathy in premature infants, (4) inhibition of free radical chain reactions of lipid peroxidation.

The signs of Vitamin E deficiency are (1) irritability, (2) edema & (3) hemolytic anaemia.

TECHNIQUE : HPLC

DECREASED IN
1. Premature & Low birth weight infants are particularly susceptible to development of Vitamin E deficiency because transplacental transfer is poor & infants have limited adipose tissue.
2. Fat malabsorption such as in Cystic fibrosis & chronic cholestasis has been known to cause neuropathy & hemolytic anaemia.
3. Rare genetic disorder Abetalipoproteinemia.

INCREASED IN
Vitamin E toxicity is usually rare & can be due to excess dietary supplementation & may cause Vitamin D & K deficiency by competing for the absorption.

This test is performed in an accredited referral laboratory.

Sample Type : *EDTA plasma*

End of Report

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M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
M.D (Pathology)

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
CALCIUM	9.2		mg/dL	8.6 - 10.0 Please note change. Source: Roche IFU.	NM-BAPTA

INTERPRETATION NOTES :

Increased level is seen in Hyperparathyroidism, Infections that cause granulomas such as tuberculosis and certain fungal and mycobacterial infections, Multiple myeloma, T cell lymphoma and certain other cancers, Metastatic bone tumor, Paget disease, Use of certain medicines such as lithium, tamoxifen, and thiazides.
Decreased level is seen in Kidney failure, Low blood level of albumin, Liver disease, Magnesium deficiency, Pancreatitis, Vitamin D deficiency

# BIOTIN (VITAMIN H)	0.4		ng/mL	0.22 - 3.00	LC-MS/MS

INTERPRETATION NOTES :

• Biotin, vitamin B7, or vitamin H, is a water-soluble vitamin.

• The vitamin plays a role in the transferring of carbon dioxide in the metabolism of fat, carbohydrate and protein by functioning as an enzyme cofactor.

• Deficiency in the vitamin may result in seborrheic dermatitis, alopecia, myalgia, hyperesthesia, and conjunctivitis.

• Disorders of biotin metabolism can be acquired or congenital.

• The lack of biotin-dependent pyruvate carboxylase, propionyl-CoA carboxylase, methylcrotonyl-CoA carboxylase, and acetylCoA carboxylase can lead to the life-threatening disorder of multiple carboxylase deficiency.

Reference:

1.Russell RM, Suter PM: Vitamin and trace mineral deficiency and excess. In: Fauci AS, Kasper DL, Braunwald E, et al. eds. Harrison's Principles of Internal Medicine. 17th ed. McGraw-Hill Book Company; 2008:441-449

2.Ball GFM: Vitamins: Their role in the human body. Oxford, Blackwell Publishing; 2004:289-299

3.McCormick DB: Riboflavin. In: Shils ME, Shike M, Ross AC, et al. Modern Nutrition in Health and Disease. 10th ed. Lippincott Williams and Wilkins; 2006:434-441

This test is performed in an accredited referral laboratory.

# COPPER	**130.21**	**H**	ug/dL	74 - 130	ICP-MS

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					

Comments : Correlation with clinical history suggested.

INTERPRETATION NOTES :

Copper Interpretation	Associated conditions
Copper also is an essential trace element that is required in enzyme systems, which in turn are responsible for countless metabolic processes required to sustain life. The major sources of excess copper are: Copper water pipes, especially when attached to a water softening system, Copper IUD's, Oral contraceptives with their estrogen content.	•Low serum copper, most often due to excess iron or zinc ingestion and infrequently due to dietary copper deficit, results in severe derangement in growth and impaired erythropoiesis. Low serum copper is also observed in hepatolenticular degeneration (Wilson disease) due to a decrease in the synthesis of ceruloplasmin and allelic variances in cellular metal ion transporters. Other disorders associated with decreased serum copper concentrations include malnutrition, hypoproteinemia, malabsorption, nephrotic syndrome & Menkes disease (kinky hair disease). •Copper Excess leads to Low Energy, Chronic Fatigue, muscle cramps, arthritis, headaches, depression, hypothyroidism.

1. Serum metal testing is used for the detection of recent exposure or poisoning with the toxic element. However, blood metal levels in healthy subjects can vary considerably with exposure to the particular metal present in the diet and in the environment.
2. It should be noted that low or within acceptable levels in serum do not always exclude that the element is uninvolved in contributing to the patient's symptoms because certain elements may be sequestered in tissues.
3. Lower metal levels in patients on follow-up imply that the toxic element exposure is reduced in the patient's immediate environment or that the body has efficiently eliminated the toxic element.
4. Free Copper is useful in diagnosis of Wilson's disease. Free Copper is estimated using following formula Free copper = Total Copper (µg/dl) – (3 X Ceruloplasmin (mg/dl)).
5. Free Copper > 25 µg/dl is observed in cases of Wilson's disease. Please update urgently as we require this for Audit purpose.

Reference –

1. Sample collection guidelines for trace elements in blood and urine. International union of pure and applied chemistry clinical chemistry division commission on toxicology working party. Pure & Appl. Chem., Vol. 67, Nos 8/9, pp. 1575-1608, 1995.
2. Nutrient & toxic elements interpretative guide, metamatrix, USA, 2011.

# MANGANESE (SERUM)	0.86		ug/L	0.3-1.04	ICP-MS

INTERPRETATION NOTES :

Reference : Rodushkin I. et al.,Pure & Applied Chem.,2001, 5, pp 51-66.

This test is performed in an accredited referral laboratory.

# SELENIUM (SERUM)	**95.67**	**H**	ug/L	74-90	ICP-MS

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
M.D (Pathology)

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P.O Box: 49527 Dubai, UAE Tel: +971 4 398 8567 reports@biosytech.ae www.biosytech.com

BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					

Comments : Sample is Rechecked to rule out analytical variability . Results require clinical correlation.

INTERPRETATION NOTES :

Selenium Serum Interpretation	Associated conditions
Selenium is vital trace element with Fish and shellfish making up as most important sources followed by meat & grains. Selenium is an antioxidant building block, which in turn are essential to neutralize free radicals in the body, thereby reducing or preventing cell membranes and DNA damage. Selenium poisoning of water systems may result whenever new agricultural runoff courses through normally dry, undeveloped lands.	•Selenium deficiency occurs as a result of sustained Total parental nutrition or restricted diets - affects physiologic systems, including endocrine and reproductive, hepatic, cardiovascular, immunological, gastrointestinal, and musculoskeletal systems. •Selenium toxicity, called selenosis, is rare in humans. Symptoms include garlic breath odour, thick brittle fingernails, dry brittle hair, red swollen skin of the hands and feet, and neurological complaints including numbness, convulsions, or paralysis.

1. Whole Blood / serum metal testing is used for the detection of recent exposure or poisoning with the toxic element. However, blood metal levels in healthy subjects can vary considerably with exposure to the particular metal present in the diet and in the environment.

2. It should be noted that low or within acceptable levels in blood / serum do not always exclude that the element is uninvolved in contributing to the patient's symptoms because certain elements may be sequestered in tissues.

3. Lower metal levels in patients on follow-up imply that the toxic element exposure is reduced in the patient's immediate environment or that the body has efficiently eliminated the toxic element.

Reference –

1. Sample collection guidelines for trace elements in blood and urine. International union of pure and applied chemistry clinical chemistry division commission on toxicology working party. Pure & Appl. Chem., Vol. 67, Nos 8/9, pp. 1575-1608, 1995.

2. Nutrient & toxic elements interpretative guide, metamatrix, USA, 2011.

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# VITAMIN K1	1.07		ng/mL	0.20-3.20	HPLC REVERSE PHASE CHROMATOGRAPHY

INTERPRETATION NOTES :

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

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M.D (Pathology)

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology

VITAMINS AND MINERALS PROFILE

1. Vitamin K is an essential, lipid-soluble vitamin that plays a vital role in the production of coagulation proteins.
2. There are 3 forms of vitamin K. Vitamin K1 (phylloquinone) isthe natural form found in green leafy vegetables, green tea, and oils such as soybean, cottonseed, canola and olive oil. It is also the commercially available synthetic form used for treatment today.
3. Vitamin K deficiency may affect any age group but is encountered most often in infancy. Infants with vitamin K deficiency which may be caused by the limited transplacental transfer of vitamin K, the low level of vitamin K in breast milk, limited neonatal liver vitamin K storage, and low neonatal colonic bacterial colonization are at risk for hemorrhagic disease of newborn.

Test Interpretation:

This test aids in estimating levels of circulating Vitamin K1 concentration. Low vitamin K1 concentrations in the serum are indicative of insufficiency and poor vitamin K1 status.

Test Limitation:

Deficiency or low Vitamin K1 levels are known due to Antibiotics (cephalosporins), Warfarin, Salicylates, Anticonvulsants, Sulfa drugs, high doses of vitamins A and E, Bileacid sequestrants (cholestyramine, colestipol), mineral oils, and orlistat weight-loss medication. It is recommended to correlate the result with complete clinical and treatment history of the same.

This test is performed in an accredited referral laboratory.

CHLORIDE (CL)	102		mmol/L	98 - 107 Please note change. Source: Roche IFU.	ISE (Indirect)

INTERPRETATION NOTES :

Increased level is seen in dehydration, with ammonium chloride administration, with renal tubular acidosis (hyperchloremic metabolic acidosis) and with excessive infusion of normal saline,hyperparathyroidism.
Decreased level with overhydration, congestive failure, syndrome of inappropriate secretion of ADH, vomiting, gastric suction, chronic respiratory acidosis, Addison disease, salt-losing nephritis, burns, metabolic alkalosis, and in some instances of diuretic therapy.

FOLIC ACID	11.4		ng/mL	Deficient: 0.35 - 3.37 Intermediate: 3.38 - 5.38 Normal: >5.38	ECLIA

INTERPRETATION NOTES :

1. Decreased levels of folic acid are seen in megaloblastic anemia, malabsorption inclusing tropical sprue, alcoholism, and inadequate dietary intake and are associated with a higher risk of fetal malformations during pregnancy.
2. In patients taking methotrexate therapy, antibodies formed may interfere with the assay.

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

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M.D (Pathology)

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
MAGNESIUM	2.5		mg/dL	1.6 - 2.6 Please note change. Source: Roche IFU.	Methylthymol blue

INTERPRETATION NOTES :

Increased magnesium levels relate mostly to patients in renal failure. Increased serum magnesium is also found with Addison disease and in pregnant patients with severe preëclampsia or eclampsia who are receiving magnesiumsulfate as an anticonvulsant. Hypermagnesemia may occur in patients using magnesium-containing cathartics. High magnesium levels are manifested by decreased reflexes, somnolence and heart block.
Decreased level or Hypomagnesia is seen in neuromuscular disorders. Hypomagnesemia is associated with hypocalcemia, hypokalemia, long-term hyperalimentation, intravenous therapy, diabetes mellitus, especially during treatment of ketoacidosis; alcoholism and other types of malnutrition; malabsorption; hyperparathyroidism; dialysis; pregnancy and hyperaldosteronism. Renal loss of magnesium occurs with cis-platinum therapy. Drugs like amphotericin toxicity will causes hypomagnesemia.

POTASSIUM (K)	4.3		mmol/L	3.5 - 5.1 Please note change. Source: Roche IFU.	ISE (Indirect)
SODIUM (NA)	138		mmol/L	136 - 145 Please note change. Source: Roche IFU.	ISE (Indirect)

INTERPRETATION NOTES :

Hypernatremia will be seen in dehydration, Cushing syndrome, central or nephrogenic diabetes insipidus with insufficient fluids, primary aldosteronism, lactic acidosis, azotemia, weight loss, nonketotic hyperosmolar coma.
Hyponatremia occurs with nephrotic syndrome, cachexia, hypoproteinemia, intravenous glucose infusion, in congestive heart failure and other clinical entities. Serum sodium is a predictor of cardiovascular mortality in patients in severe congestive heart failure. Addison disease, hypopituitarism, cirrhosis, hypertriglyceridemia and psychogenic polydipsia.

ZINC (SERUM)	**121**	**H**	ug/dL	70 - 114	Endpoint

INTERPRETATION NOTES :

1. Whole Blood / Serum metal testing is used for the detection of recent exposure or poisoning with the toxic element. However, blood metal levels in healthy subjects can vary considerably with exposure to the particular metal present in the diet and in the environment.
2. It should be noted that low or within acceptable levels in blood / Serum do not always exclude that the element is uninvolved in contributing to the patient's symptoms because certain elements may be sequestered in tissues.
3. Lower metal levels in patients on follow-up imply that the toxic element exposure is reduced in the patient's immediate environment or that the body has efficiently eliminated the toxic element.
4. Zinc is vital trace element required for normal healing of wounds and normal immune function. Zinc levels lowered with systemic infections & inflammatory disorders, oral contraceptives and pregnancy.
5. Zinc exposure can occur from occupational exposure related to mining and metallurgic industries and use of commercial products containing zinc

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

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M.D (Pathology)

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BIOCHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
(e.g. Zinc containing shampoos, Multivitamins)					

Associated conditions:

1. Zinc deficiency - symptoms include depressed growth, teratogenesis, poor carbohydrate metabolism, altered cognition, poor immune function, alopecia, impotence,eye and skin lesions, and diarrhoea.
2. Zinc excess is not of major clinical concern, however elevated zinc concentrations may interfere with copper absorption.

Reference:

1. Sample collection guidelines for trace elements in blood and urine. International union of pure and applied chemistry clinical chemistry division commission on toxicology working party. Pure & Appl. Chem., Vol. 67, Nos 8/9, pp. 1575-1608, 1995.
2. Nutrient & toxic elements interpretative guide, metal matrix, USA, 2011.

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

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M.D (Pathology)

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IMMUNOLOGY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
VITAMIN D, 25-OH (TOTAL)	25		ng/mL	Deficiency: <20 Insufficiency: 20 - <30 Sufficiency: 30 - 80 Toxicity: >80 Please note change. Source: Roche IFU.	ECLIA

INTERPRETATION NOTES :

Vit D (25 – OH) is the sum of Vit D2 (25 – OH) and Vit D3 (25 – OH). In normal persons not taking external supplements - Vit D3 comprises approximately 90 % of the total.

25 hydroxy (25–OH) vitamin D3 or calcidiol is the storage form of vitamin D3. Deficiency is associated with osteoporosis, multiple sclerosis, and rheumatoid arthritis, and mood disorders. Both Vitamin D2 and Vitamin D3 are converted to 25–OH vitamin D3 in the liver. 25 hydroxy vitamin D3 circulates to the kidney where it is converted to 1, 25 hydroxy vitamins D3 or calcitriol, the functional form of the vitamin.

Calcitriol is vital to calcium regulation and low serum calcium causes release of parathormone which converts 25–OH vitamin D3 to 1, 25 –OH vitamin D3 which then triggers osteolysis releasing calcium into the bloodstream.

Sample Type : *Serum*

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Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
M.D (Pathology)

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REFERRED

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
# Beta Carotene	227.8	H	ug/dL	48.00-200.00	Spectrophotometry

Comments : Kindly correlate clinically.

INTERPRETATION NOTES :

This test is performed in an accredited referral laboratory.

Sample Type : *OTHER SPECIMEN*

End of Report

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M.D (Pathology)
Pathologist

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M.D (Pathology)

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CLINICAL CHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					
# Chromium	2.85		ug/L	0.7-28	ICP-MS

INTERPRETATION NOTES :

Chromium Interpretation	Associated conditions
Chromium is a naturally occurring element in rocks, animals, plants, soil, and volcanic dust and gases. Thus, the general population is exposed to chromium (generally chromium [III]) through food, drinking water, and inhaling air containing the chemical. Chromium (III) is essential to normal glucose, protein, and fat metabolism and is thus an essential dietary element. Our body has several detoxification systems for reducing chromium (VI) to chromium (III). Excess Chromium exposure may occur commonly in industrial processes like manufacture of cars, glass, pottery and linoleum which involves air emissions of chromium in the form of small particles or aerosols.	Overexposure to chromium may cause gastrointestinal symptoms, such as diarrhoea and vomiting, severe water- electrolyte disorders, increased blood acidity and body tissues (acidosis). Lesions on the kidneys, liver, and muscular layer of the heart (myocardium) may also develop. Carcinogenic effects causing lung, respiratory & renal cancers.

1. Whole Blood metal testing is used for the detection of recent exposure or poisoning with the toxic element. However, blood metal levels in healthy subjects can vary considerably with exposure to the particular metal present in the diet and in the environment.
2. It should be noted that low or within acceptable levels in the blood do not always exclude that the element is uninvolved in contributing to the patient's symptoms because certain elements may be sequestered in tissues.
3. Lower metal levels in patients on follow-up imply that the toxic element exposure is reduced in the patient's immediate environment or that the body has efficiently eliminated the toxic element.

Reference –

1. Sample collection guidelines for trace elements in blood and urine. International union of pure and applied chemistry clinical chemistry division commission on toxicology working party. Pure & Appl. Chem., Vol. 67, Nos 8/9, pp. 1575-1608, 1995.
2. Nutrient & toxic elements interpretative guide, metal matrix, USA, 2011.

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# VITAMIN B2 (RIBOFLAVIN)	270.85		ug/L	180 - 295	HPLC

INTERPRETATION NOTES :
CLINICAL UTILITY:

1. Ariboflavinosis (dietary deficiency of riboflavin causes a syndrome characterized by sore throat, hyperemia, and edema of the pharyngeal and oral mucous membranes, cheilitis, angular stomatitis, glossitis, corneal vascularization, dyssebacia with dermatitis, and normochromic, normocytic anemia.
2. Severe deficiency can diminish conversion of Vitamin B6 to its coenzyme and curtail conversion of tryptophan to niacin.
3. During recent years, deficiencies of riboflavin and vitamin B6 have been suggested as potentiating factors for chronic progressive nerve compression disorders, such as carpal tunnel syndrome.

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

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M.D (Pathology)

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CLINICAL CHEMISTRY

Test	Result	Flag	Unit	Reference Range	Methodology
VITAMINS AND MINERALS PROFILE					

This test is performed in an accredited referral laboratory.

Sample Type : *EDTA Whole Blood*

End of Report

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
M.D (Pathology)

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REFERRED

Final Report

Primary Materials

Serum

analyzed specimen

	Result	Unit	Ref.-Range
Serum (s)			

Clinical chemistry

	Result	Unit	Ref.-Range
Fluoride (serum)(**)	<0.05	mg/l	<= 0,10
	therap. range: up to 0.44 mg/l		

End of Report

Dr. Adley Mark Fernandes
M.D (Pathology)
Pathologist

Dr. Vyoma V Shah
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